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2024/2025 SEMESTER 2 – PROJECT

Course : Diploma in AI and Data Engineering

Module : EGT216 - Natural Language Processing

Instructions to students:

1. Plagiarism

Nanyang Polytechnic (NYP) takes plagiarism seriously and regards it as a form of academic dishonesty. Students who violate the NYP Academic Integrity Policy of NYP by committing plagiarism will be subject to disciplinary action.

Learners will face disciplinary action if they provide or receive unauthorized assistance, such as:

- Copying another learner's answer
- Providing answers for other learners
- Falsifying data, information or quotations
- Asking another person to do the work you are supposed to do
- Taking and using ideas, words or work of others in whole or in part and passing it off as your own work without citing the original source (including AI resource)

Please refer to the NYP Academic Integrity Policy on the NYP website

Plagiarism is taking the work or ideas of another person, whether in whole or in part, and presenting it as one's own work without acknowledging the original source.

Sharing your work with other learners is considered plagiarism. ALL those involved, including those who share their work, will be penalised in accordance with the NYP Academic Integrity Policy.

Cite and reference ALL sources you have used in your report in APA style.

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2. Copyright Infringement

It is equally important that you avoid copyright infringements and name the authors of the images you use. When searching for images, make sure that you have obtained the necessary permissions or licenses to use the images in your work. Using copyrighted images without proper permission is a direct violation of intellectual property rights. You must use copyright-free or Creative Commons licensed image databases, as this will allow you to obtain imagery in an ethical manner while adhering to the principles of academic integrity.

3. Late Submission

The dateline to submit your project report and code is **February 23, 2025**. Any submission received after the specified due date and time will be considered late.

If you pass the assignment, a 50% penalty will be applied to your base score. If you fail the assignment, you will receive a failing grade.

Regardless of whether you pass or fail, any submission made more than 5 calendar days after the due date will result in a score of zero.

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1. Project Scenario

Companies often receive a large number of resumes for each job posting, requiring them to use specialized systems to screen applicants.

In a competitive job market, it is important for individuals to write resumes that clearly highlight their qualifications and stand out to recruiters or automated systems. A good CV presents skills, experience, and achievements in a clear and simple way, making it easier for both people and machines to identify the best candidates.

You are an AI engineer at company XYZ and have been assigned the task of using NLP algorithms and models to classify and search resumes. You should explore various methods for text pre-processing, text vectorization, and NLP modeling.

2. Learning Outcomes

This assignment will assess the attainment of the following learning outcomes:

- Explain supervised and unsupervised NLP algorithms that can be deployed to gain insights from text in engineering applications.
- Identify traditional or deep-learning-based NLP algorithms to automate engineering processes based on user requirements.
- Develop insights from text data for sense-making and decision-making using text classification algorithms.
- Evaluate the effectiveness of NLP applications using standard performance metrics.

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3. Timeline

Week 15

- Briefing on the objectives and requirements of the project.
- Create code to process your given dataset using text pre-processing techniques.

Week 16

- Create code to retrieve the top 10 resumes most similar to your CV from the dataset using most suitable text vectorization technique.
- Provide an update on Week 15 task.
- Submit cover letter and resume.

Week 17

- Create code to classify the resumes in your given dataset using BERT model.
- Provide an update on the Week 16 task.

Week 18

- Optimize and enhance the overall code to improve the performance of the classification process.
- Provide an update on the Week 17 task.

Week 19

- Submit the project report and the code.